

When Do Uncertain Dates Change Retrieval? Exact Certificates for Temporal Evidence

Anonymous ACL submission

Abstract

Sources often date historical changes only to a month or year. Choosing an exact date can change the evidence retrieved for a question. We ask when resolving these dates can change the selected evidence. Our method represents each stated change by a date range and records which claims can replace others. At most two source events, called witnesses, summarize each claim when event dates vary independently. These witnesses identify evidence that could apply and evidence that must apply within a requested period. An exact test then certifies an unchanged context or returns two valid timelines with different selected claims. The test compares two filtered versions of a fixed ranking, without computing every attainable context. The same criterion supports known event order and different query meanings through an exact constraint solver. On 3,178 TimeQA questions, the automatic adapter reaches 309 contexts containing modeled evidence. It certifies 245 as stable and verifies changes for the remaining 64. Replacement changes 34 of 73 contexts in a separate source-derived challenge. On controlled dependent dates, screening removes 59.79% of the claim checks made by an otherwise identical solver. The resulting interface identifies the source dates that can change a reader's evidence.

1 Introduction

Historical questions often ask about a period, while sources give incomplete event dates. A source might report an appointment in September without identifying its day. Choosing the first or last day can change which evidence reaches a reader. Yet resolving every uncertain date wastes effort when those dates cannot change the selected context. We ask a practical question: *When does date uncertainty actually change retrieval?*

This problem arises before answer generation. Retrieval-augmented systems select passages be-

fore generating answers (Lewis et al., 2020). Suppose a source states that a director's term ended in 2005. The unknown departure day does not affect a question about 2004. For a question about June 2005, it can change whether that tenure remains eligible. Even then, the uncertainty matters only if the excerpt reaches the reader's selected context. The system needs a test that distinguishes these cases before requesting more precise dates.

Temporal memories already preserve histories and invalidate replaced facts (Rasmussen et al., 2025; Zerhoubi et al., 2026). Interval retrievers also handle fuzzy dates (Chen et al., 2026; Wang et al., 2026). We study which retrieval decisions can be settled before completing those dates.

Each claim receives a range containing its unknown start date. A directed decision specifies which other claims can replace it. We compile these inputs into at most two source witnesses per claim. The first determines when support becomes impossible. The second determines when guaranteed support can fail. They answer every point or period query without enumerating timelines. *Possible support* means that a claim applies somewhere in the requested period under at least one allowed timeline. *Guaranteed support* means that it applies somewhere in that period under every allowed timeline.

The resulting masks also decide whether uncertainty changes retrieval. Fix a relevance ranking and a budget of k claims. If its first k possible claims are guaranteed, every timeline returns the same context. Otherwise, we identify a selected claim whose membership changes across two explicit timelines. These timelines can be replayed through the retrieval pipeline. They identify selected support decisions that further date evidence must resolve.

Our contributions connect three levels of the problem. First, two direct witnesses give exact period-support formulas under independent date

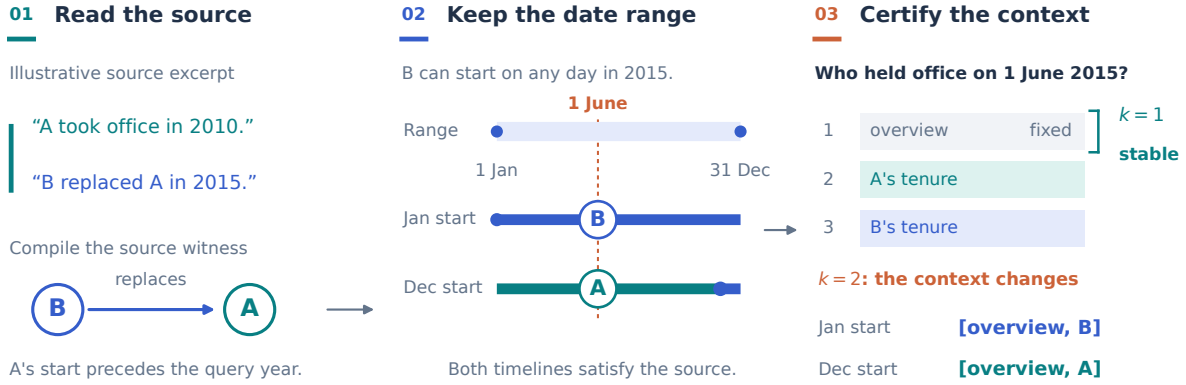


Figure 1: From source dates to a context decision. A stated replacement in 2015 can occur before or after a June query. The fixed overview excerpt gives a stable one-item context. A second slot exposes the uncertain tenure. The system returns both valid timelines and their different excerpts. This example illustrates the interface.

bounds. Compilation takes linear time in the events and supplied replacement pairs. Second, an exact context criterion separates stability from the complete attainable context envelope. It holds for any nonempty family of feasible timelines under a fixed relevance ranking. A constraint solver extends the interface to known event order and three query predicates. Third, complementary evaluations measure source coverage, replacement behavior, and response sensitivity. The automatic TimeQA track measures the adapter’s reach. A separate source-derived challenge directly tests replacement between claims. Controlled models test dependent dates and distinguish the two witnesses.

2 From Sources to Temporal Evidence

An appointment in 2001 bounds one event. A tenure from 2001 to 2005 supplies separate start and end events.

Occurrence precision. A day gives a point on the time axis. A month or year bounds an occurrence within that calendar period. Expressions such as *around 1713* require additional evidence before they supply finite bounds. We preserve the literal date expression and its source span. The query adapter applies the same convention to the visible question. A question about a whole year therefore remains a period query. Our predicate asks whether a claim applies *somewhere* within that period.

Replacement and coexistence. A directed replacement decision connects values of the same subject and role. For office succession, the organization and office define the slot; people supply its values. For affiliation, a new organization can coex-

ist with an older affiliation. The Chicago Stadium passage in TimeQA, for example, gives overlapping tenancies for two teams (Chen et al., 2021).

Explicit ends. A stated departure supplies an end event targeting its own claim. The compiler treats this hidden event as a direct replacement. The independent compiler requires the end’s lower bound to exceed the start’s upper bound. Section 4.5 handles overlapping bounds through lifetime constraints. This reduction preserves gaps between tenures. It also handles a departure without a named successor. Appendix E proves equivalence to the corresponding half-open lifetime.

Evidence units. A paragraph can contain several states, so temporal selection operates on source excerpts. Each modeled claim preserves exact character offsets and its supporting sentence for audit. Uncovered text remains available through a shared fallback policy. An incomplete temporal parse retains the complete source text. Readers receive original excerpts, date bounds, and support tags.

Binding dates to claims. The source adapter requires an explicit local subject or a unique paragraph-level antecedent. It links a paragraph-level departure only when the target is unique. Recognized bounded states with unresolved endings remain fallback text.

Support tags. Modeled claims are guaranteed, unresolved, or unsupported for the requested period. Unresolved claims have possible support without guaranteed support. Unmodeled excerpts carry a separate *unknown* tag. Their fixed fallback eligibility lets us isolate uncertainty in modeled dates.

3 Dates as Constraints

Occurrence dates and replacement. Each claim c has text and an unknown occurrence date x_c . Its source bounds this date by two inclusive dates, $\ell_c \leq x_c \leq u_c$. Dates use a common real-valued time axis. These bounds describe uncertainty about an occurrence, rather than its duration. Each timeline independently chooses one date per range: $\Omega = \prod_c [\ell_c, u_c]$. A fixed text gate $G(d, c)$ records whether d can replace c . The gate need not be transitive. It excludes self-pairs: $G(c, c) = 0$. It must distinguish replacement from coexistence. The compiler preserves its decisions across date assignments.

For assignment x , claim c is active at date t when

$$A_x(c, t) = 1[x_c \leq t \wedge \nexists d : G(d, c), x_c < x_d \leq t]. \quad (1)$$

Equal dates do not establish replacement order. A question specifies an inclusive date window $Q = [a, b]$. Point questions set $a = b$. A claim supports the window when it is active somewhere inside it:

$$S_x(c, Q) = 1[\exists t \in Q : A_x(c, t) = 1]. \quad (2)$$

Possible support requires one allowed assignment with $S_x(c, Q) = 1$. Guaranteed support requires this condition in every allowed assignment. Write $P_Q(c)$ and $H_Q(c)$ for these two Boolean support decisions.

Two direct witnesses. One witness can force retirement despite date uncertainty. Another can permit retirement under a particular assignment. Their boundaries are

$$p_c = \min\{u_d : G(d, c), \ell_d > u_c\}, \quad (3)$$

$$r_c = \min\{\ell_d : G(d, c), u_d > \ell_c\}. \quad (4)$$

An empty minimum is $+\infty$. We store one witness for each finite boundary. The representation uses at most two witness records per claim. Two records are necessary in the worst case among direct witness subsets.

Theorem 1 (Exact window support). *For every claim and inclusive query window,*

$$P_Q(c) \iff \ell_c \leq b \wedge a < p_c, \quad (5)$$

$$H_Q(c) \iff u_c \leq b \wedge (a \leq \ell_c \vee a < r_c). \quad (6)$$

The first condition gives the smallest filter that never removes potentially supported evidence. The

second gives the largest filter whose retained claims always have support. Each decision takes constant time after compilation. Appendix A gives the proofs.

Guaranteed support allows its supporting time to differ across timelines. For example, let $x_c \in [0, 10]$ and let a replacement start at time 5. Claim c always contributes somewhere in $[0, 10]$, either before the replacement or after its own later start. Yet no single point supports c across all timelines. A certainly active interval therefore cannot replace the window test.

Corollary 1 (Refinement and local influence). *Narrowing date bounds can only shrink possible support and expand guaranteed support. If two gates differ on h directed pairs, either support mask changes for at most h claims. With fixed rankings, their filtered top- k contexts differ in at most $2 \min(k, h)$ items.*

Each directed pair affects only its target claim's certificates. When all date bounds are exact, both filters recover earliest direct replacement. The formulas therefore extend the same policy without choosing arbitrary dates.

4 Certifying the Reader's Context

4.1 Compile once, query each claim in constant time

One scan updates both boundary minima, breaking ties by stable source identifiers. Figure 2 shows why the two minima can require different witnesses.

Let n count modeled events, including explicit ends, and m count accepted pairs. Compilation takes $O(n + m)$ time and stores $O(n)$ certificate records. Pair discovery remains the source adapter's responsibility. The reference complete scan makes $n(n - 1)$ gate calls. An index can replace this scan when it preserves every accepted pair.

4.2 An exact stability test

Let π_q be a complete relevance ranking for question q , including all tie breaks. For a set S , write $\text{Top}_k(S)$ for its first k ranked claims. Let $\mathcal{W}$ be any nonempty family of feasible timelines. Define $P(Q)$ and $H(Q)$ by possible and guaranteed support over $\mathcal{W}$. Only displayable claims enter these support sets. The fast compiler uses $\mathcal{W} = \Omega$. A realized timeline x supplies a support set $S_x(Q)$.

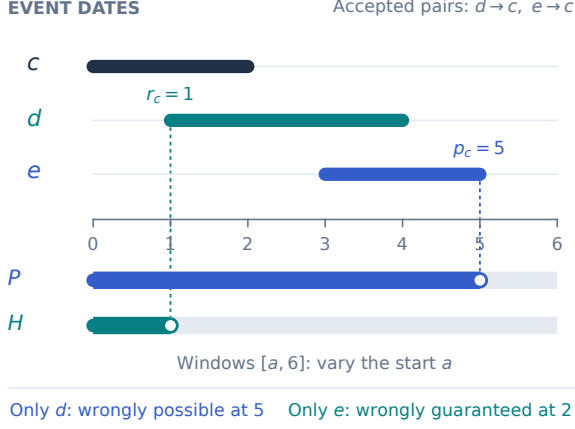


Figure 2: Two witnesses control different boundaries. Witness d supplies $r_c = 1$, while e supplies $p_c = 5$. The strips show support for windows $[a, 6]$. Open endpoints exclude equality. Removing e wrongly permits point 5; removing d wrongly guarantees point 2.

The reader's selected claim context is

$$R_x(q) = \text{Top}_k(S_x(Q)). \quad (7)$$

Filtering preserves the ranking and occurs before the budget.

Theorem 2 (Exact context stability). *For any nonempty family of feasible timelines, every timeline returns the same context exactly when*

$$\text{Top}_k(P(Q)) = \text{Top}_k(H(Q)). \quad (8)$$

Equivalently, every selected possible claim has guaranteed support.

Proof. Set $C = \text{Top}_k(P(Q))$. If $C \subseteq H(Q)$, every world retains C and selects the same context. Otherwise, choose $c \in C \setminus H(Q)$. Some world supports c , while another does not. At most $k - 1$ possible claims outrank c . Every world supporting c therefore selects it. The two worlds return different contexts. For $k = 0$, every context is empty. $\square$

The theorem depends on fixed ranking and exact support masks. It applies to dependent dates and other support predicates. Only the support solver changes. The test uses two filtered rankings without enumerating attainable contexts. Let F contain the fixed fallback excerpts. Apply the test to $P(Q) \cup F$ and $H(Q) \cup F$. Their inclusion remains constant while modeled dates vary. Claim selection precedes source rendering and any word-budget truncation. We report rendered-context agreement separately.

Choose a stable context budget. Let r be the position of the first unresolved claim in the ranking of $P(Q) \cup F$. Every budget $k < r$ returns a fixed context, while every $k \geq r$ permits a context change. If no unresolved claim exists, every budget is stable. The claim at position r supplies a counterexample for every larger budget. The system reports this stable prefix alongside the source dates requiring refinement.

Stop once the context is full. The implementation evaluates certificates lazily during selection. It scans ranked candidates until k possible excerpts fill the context. It checks guaranteed support only for selected modeled claims. Fixed fallback excerpts remain eligible without a temporal lookup. If selection examines j candidates, selection and certification take $O(j)$ time. Other dates need no evaluation for this decision.

4.3 Constructing and resolving an unstable context

Let $U = \text{Top}_k(P(Q) \cup F) \setminus (H(Q) \cup F)$. These claims have unresolved support and enter the possible context. Their status determines whether date uncertainty can change retrieval.

Theorem 3 (Constructive instability and refinement). *Under independent date bounds, if $U \neq \emptyset$, we construct two complete date assignments with different contexts in $O(n)$ time. Here, n counts the event dates. Any stabilizing date refinement makes each claim in U guaranteed or impossible.*

Each $c \in U$ has fewer than k possible claims above it. Any timeline supporting c therefore selects it. We place each accepted replacement of c no later than c or after a supporting query time. A second timeline places c after the query, or a direct witness by the query start. Date refinement can only remove possible claims. Thus a remaining possible pivot cannot leave the possible context through rank changes. It must become guaranteed. The assignments use only event bounds and the compact certificate. Let N count ranked excerpts, including fallback text. Full context replay takes $O(n + m + N)$ time with the accepted pair list. Appendix F gives the construction and proof.

The witnesses locate the start or replacement date requiring refinement. New unresolved claims can enter after current pivots disappear, so refinement recomputes the frontier.

4.4 A tractability boundary

Possible support concerns one claim at a time. Possible context membership also depends on which higher-ranked claims coexist. We ask whether a given claim enters the context under *any* allowed timeline.

Theorem 4 (Possible context membership). *Possible context membership is NP-hard when k is input; this result does not cover the fixed budget $k = 5$.*

The reduction uses a point query and a bipartite gate with one edge layer. Appendix H gives the proof.

The appendix also gives a reduction with every start no later than the query. Our stability test instead needs one unresolved claim inside the possible prefix. Its rank guarantees selection whenever it has support. The test needs only individual support, even when shared events couple claim eligibility. Under independent bounds, this certifies stability without computing every attainable context member.

4.5 Enforce known order only where it matters

A source can state that one appointment follows another without identifying either exact date. We represent these dependencies as difference constraints, following temporal constraint networks (Dechter et al., 1991). This extension uses integer calendar days. For example, $x_i - x_j \leq -1$ states that event i precedes event j . A departure date z_c satisfies $x_c - z_c \leq -1$ and supplies a replacement pair targeting c . This permits overlapping start and end bounds while retaining only valid lifetimes. We check that the resulting feasible family $\Omega_D \subseteq \Omega$ is nonempty.

Select the query predicate. The interface distinguishes three temporal meanings. Query meaning is an explicit input. Overlap asks whether the claim applies somewhere in $[a, b]$. Throughout asks whether it applies at every date in that window. Appointment asks whether its start belongs to the window. Let $D(c) = \{d : G(d, c) = 1\}$. The first two predicates have the common form

$$x_c \leq s \wedge \bigwedge_{d \in D(c)} (x_d \leq x_c \vee x_d > t), \quad (9)$$

where $(s, t) = (b, a)$ for overlap and (a, b) for throughout. Appointment instead requires $a \leq x_c \leq b$.

An exact oracle searches feasible branches of Equation 9 and its negation. It returns supporting and excluding timelines when they exist. Possible support requires the former; guaranteed support excludes the latter. Each branch uses standard difference-constraint feasibility. Appendix I proves completeness and gives the branching cost. The context criterion in Theorem 2 then applies without changing the ranking.

Screen first, solve selectively. For overlap, let P_0, H_0 denote the independent compiler’s support sets. Let P_D, H_D denote their exact constrained counterparts. Restricting the timeline family gives

$$H_0 \subseteq H_D \subseteq P_D \subseteq P_0. \quad (10)$$

Equal product contexts therefore certify the constrained context without any query-level oracle call. Otherwise, scan the fixed ranking and skip claims outside P_0 . Accept claims in H_0 as guaranteed. Query the exact oracle for remaining candidates until k possible claims fill the context. Fixed fallback excerpts extend this scan by entering every support set without an oracle call. The resulting context is $\text{Top}_k(P_D \cup F)$. An unresolved selected claim supplies the two replayable timelines.

5 Evaluation Protocol

We evaluate replacement behavior, dependent dates, source coverage, and answer sensitivity. The natural-source experiments use original TimeQA paragraphs (Chen et al., 2021). Source extraction receives article titles, paragraph text, identifiers, and paragraph order. Retrieval receives only the visible question. Answers and annotated spans enter scoring only.

Replacement challenge. A fixed source-hash sample selects paragraphs containing explicit succession cues. Two model-assisted reviews check claim boundaries, dates, and directed replacement links before temporal evaluation. Accepted graphs define a source-derived mechanism diagnostic. Each successor supplies point queries at its earliest, midpoint, and latest dates. Source order fixes the ranking; budgets are one and five excerpts. Both replacement and interval controls enforce every known temporal order. Appendix J gives all sampled paragraphs and review decisions.

Controlled constraints. We generate 120 feasible models with five visible claims and up to six

event dates. They cover precedence, lifetime constraints, and their combination. Twenty further models are inconsistent. Enumeration over seven integer dates supplies an independent exact reference. The evaluation crosses 28 windows, three query predicates, and three budgets. A separate grid isolates cases needing both direct witnesses.

Automatic source extraction. A fixed hash selects 176 development questions from 50 articles for ranking calibration. Separate source inspections guide extraction rules and their semantic audit. The revised source adapter remains fixed throughout evaluation. Automatic TimeQA retrieval uses overlap support. We evaluate 2,348 template questions from 644 articles and 830 human-written questions from 251 articles. Both tracks search their complete source pools. These contain 32,332 and 33,493 paragraphs, respectively. The appendix records earlier inspected outcomes, calibration exclusions, and all input hashes.

Fixed ranking and matched controls. The selected ranking links the longest public article title appearing in the question. It removes matched title words before BM25 scoring (Robertson et al., 1995). Linked excerpts receive priority; all other corpus excerpts remain in the complete ranking. Six ranking variants compete on the pilot before confirmation. Every temporal policy shares source units, bounds, replacement pairs, ranking, and tie breaks. We compare earliest, midpoint, and latest date completion against possible and guaranteed support. The interval control tests start and explicit-end overlap, ignoring cross-claim replacement. The no-filter control retains all excerpts. Separate ranking controls use original BM25 and latest-year reranking within twenty lexical candidates. We also apply that recency rule to the selected title-aware ranking.

Evidence budgets and measurements. Each context holds five source excerpts and at most 768 source-text words. Every method retains unmodified text under the same fallback policy. We average each question’s fraction of annotated spans fully contained in its shown excerpts. A parent paragraph match alone earns no span credit. We report stability separately for contexts with modeled claims and contexts containing only fallback text. Every unstable context receives two complete date assignments with different selections. An independent evaluator replays those assignments. Con-

fidence intervals use 2,000 paired article-cluster resamples, with seed 20261006.

Answer generation. We run Qwen2.5-3B-Instruct with official Q4_K_M weights and a pinned CPU runtime (Qwen Team, 2024). A fixed hash selects 60 human questions from 55 articles before generation. Ten methods yield 600 logical conditions. Identical prompts share one generation, with every question–method membership retained. Temperature is zero, seed is 1729, and the output limit is 512 tokens. JSON-schema constrained decoding enforces the response structure. The prompt requests all supported answers and cited excerpt numbers in JSON. The prompt and parser remain fixed across methods. An eleven-question diagnostic reuses a historical cohort selected for context disagreement. The primary and diagnostic cohorts produce 194 and 40 unique requests, respectively. Invalid responses contribute empty answers and remain in every denominator. We score exact annotated answer sets and strict set F1. The release contains complete prompts, raw outputs, timing, and request hashes.

6 Results

6.1 Replacement changes temporal eligibility

The source-derived challenge contains twenty accepted paragraphs from eighteen articles. Its 45 claims have 33 directed replacement links. Twenty-five links record consecutive replacements. Eight links follow complete ordered succession lists. Both source reviews check every accepted graph before certificate evaluation. The frozen boundary schedule yields 73 point queries.

Both policies retain identical date bounds, explicit ends, and known source orders. With five excerpts, replacement changes 34 contexts relative to interval overlap. With one excerpt, omitting replacement marks 42 unstable contexts as stable. Every unstable decision has two order-feasible timelines with different literal excerpts. Table 1 reports both budgets.

Known order also sharpens the source decisions. A Levadia passage places Bondarenko’s appointment in 2001 and Rautiainen’s succession in November 2001. Enforcing their stated order resolves one additional stable query at each budget. Under independent dates, the fast compiler matches all 388 exact support judgments. Appendix J records sampling, reviewed annotations, query construction, and complete witness replay.

k	Queries	Changed	Missed	Stable
1	73	33	42	25
5	73	34	0	25

Table 1: Source-derived replacement challenge. Changed counts possible contexts differing from interval control. Missed counts replacement-sensitive contexts that interval control marks stable. Stable counts certified contexts with replacement. Both policies enforce known source order.

Budget	Lazy exact	Hybrid	Saved (%)
1	4,361	1,645	62.28
3	12,019	4,849	59.66
5	16,800	6,849	59.23
All	33,180	13,343	59.79

Table 2: Exact claim queries on 120 constrained models and 28 windows. Both methods stop after filling the possible context. Each budget covers 3,360 requests. Their contexts and stability decisions agree throughout.

6.2 Constraints resolve avoidable uncertainty

Exhaustive world enumeration confirms all 100,800 support decisions and 30,240 fixed-ranking context decisions. The oracle rejects all twenty inconsistent models. Across the three predicates, constraints certify 929 contexts left unresolved by independent bounds. These cases include overlapping lifetime bounds and uncertain event order.

For overlap queries, the hybrid first certifies 3,001 contexts without query-level constraint solving. Selective solving recovers 312 additional stable contexts. It returns two replayed timelines for each of the remaining 6,767 unstable contexts. Table 2 compares it with a lazy exact solver using the same ranking and stopping rule. The hybrid uses 59.79% fewer exact claim queries and 55.67% fewer feasibility calls. These counts measure solver work after ranking.

A separate grid tests 476 configurations with two distinct supplying witnesses. The complete compiler matches all 13,328 claim-window decisions. Each single-witness subset fails in every configuration. Removing the forced-retirement witness wrongly includes 1,750 possible cases. Removing the optional-retirement witness wrongly includes 5,576 guaranteed cases.

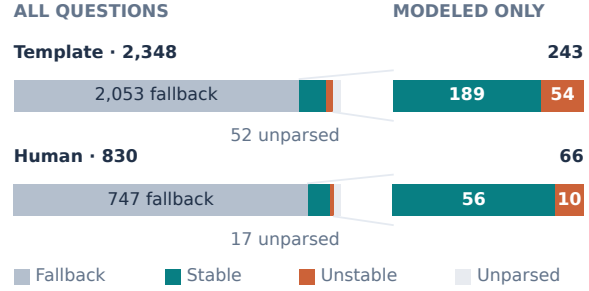


Figure 3: Coverage before conditional stability. The left bars show every question. The right bars expand contexts containing modeled claims. Gray fallback excerpts have fixed eligibility. Stable and unstable counts apply only to the modeled subsets.

6.3 Source coverage controls the automatic track

Figure 3 reports every evaluated question. The automatic adapter reaches modeled contexts for 309 of 3,178 questions, or 9.72%. It finds 245 stable contexts and 64 unstable contexts within this group. Stable means unchanged selected excerpts under the supplied date model. The other questions yield 2,800 parsed all-fallback contexts and 69 unparsed windows. Fallback eligibility stays fixed and contributes no temporal certificate.

The adapter models 493 development claims and 464 test claims. They occupy 1.49% and 1.36% of indexed source units. A model-assisted source audit accepts 54 of sixty event interpretations. Grounding blocks both learned extraction pilots. Two learned extraction variants return 87 records from six shared calibration paragraphs. None passes both literal grounding and holder-date checks. The deterministic adapter therefore requires explicit supporting spans before issuing a temporal certificate. Appendix G retains every source judgment and failed extraction output.

Independent replay verifies all 128 date assignments for the 64 unstable contexts. Each pair changes the rendered source excerpts. Forty-one exclusions use explicit retirement dates; twenty-three use late starts. For example, a human question asks which title Peter Beattie held in February 1998. The source dates his premiership from 1998 to 2007. Starts on February 1 and December 31 produce different contexts while respecting that source precision. The automatic track exercises these interval boundaries.

Method	Primary	Diagnostic
Original + year	25.00	18.18
Title-aware	36.11	9.09
Title-aware + year	33.33	24.24
Possible support	36.11	9.09
Guaranteed support	36.11	18.18

Table 3: Strict answer-set F1 (%). Temporal policies use title-aware BM25 without the year bonus.

6.4 Certificates compose with the relevance ranker

The selected title-aware ranking reaches 67.61% template span recall and 58.11% human span recall. Its gains over original BM25 with recency are 14.34 and 5.00 percentage points. Their paired intervals are [11.86, 16.77] and [0.69, 9.10]. Adding recency to the selected ranker reaches 64.26% human recall. These gains belong to relevance ranking. Possible support and the interval control select identical excerpts on the automatic track. The appendix gives all ten matched retrieval rows.

Certificates also apply to all six prespecified ranking variants. Every one of 18,654 query–ranking decisions matches full-mask selection. Under the strongest human retrieval control, 102 of 120 modeled contexts remain stable. Larger excerpt budgets expose additional uncertain claims. The appendix reports every budget comparison.

6.5 Answer sensitivity under structured decoding

All 234 responses pass the unchanged parser, covering every primary and diagnostic condition. Table 3 reports strict answer-set F1. The title-aware ranking gains 11.11 points over original BM25 with recency, with interval [3.33, 19.44]. This difference follows the relevance ranking. Possible and guaranteed support share every primary context and produce identical answers.

Seven diagnostic questions have different possible and guaranteed contexts. Four produce different answer sets. Guaranteed support gains 9.09 F1 points, with interval [0.00, 30.00]. Partial strict matches now separate F1 from exact match. Appendix K gives cardinality strata, every changed answer, and the complete decoding comparison.

7 Related Work

Temporal memory and replacement. Graphiti closes earlier facts through direct contradictory evidence (Rasmussen et al., 2025; Graphiti contribu-

tors, 2026). MemStrata applies deterministic supersession to structured keys (Yadav, 2026). ERASE edits stored fact histories as new documents arrive (Li et al., 2025). NuggetIndex attaches provenance and temporal states to atomic records (Zerhoubi et al., 2026).

Temporal retrieval. Re³ combines fact grouping, conflict filtering, and temporal relevance (Cao et al., 2026). DeFuzzRAG adapts granularity and filters document scopes by query overlap (Chen et al., 2026). IA-RAG represents intervals and refines uncertain event dates (Wang et al., 2026). TA-RAG retrieves evidence across timelines (Lau et al., 2025). Our certificates distinguish possible and guaranteed support across every permitted occurrence date.

Uncertain temporal data. Anselma et al. (2013) study valid-time indeterminacy and compact representations. Fan et al. (2012) study data currency without reliable timestamps. Amarilli et al. (2017b) study possible ordered answers and query-result certainty.

Uncertain ranked answers. Hua et al. (2008) study top-*k* membership probabilities under uncertain record membership. Amarilli et al. (2017a) study unknown ranking values constrained by a partial order. Feng et al. (2023) bound ranking results and SQL window queries. We keep relevance scores fixed and derive uncertain eligibility from direct temporal replacement.

Temporal evaluation. TimeQA supplies natural passages and questions about dated histories (Chen et al., 2021). TempLAMA evaluates factual answers at yearly resolution (Dhingra et al., 2022). StreamingQA and FreshLLMs study changing information access (Liška et al., 2022; Vu et al., 2024). FRESCO distinguishes obsolete evidence from current but insufficient passages (An et al., 2026).

8 Conclusion

Two temporal witnesses make date uncertainty actionable at the point of evidence selection. The retriever returns a stable context, or explicit timelines and support decisions that explain why its evidence can change.

Limitations

The context guarantee assumes a fixed relevance ranking and exact support masks. A ranking that depends on uncertain dates can change even when claim eligibility stays fixed. Appendix I gives a concrete counterexample and controlled comparisons. The fast compiler uses independent date bounds. The constraint extension enforces known event order and valid lifetimes, including overlapping start and end bounds. Its exact search can branch exponentially with replacement degree. The implemented predicates cover overlap, uninterrupted validity, and appointment within a window. More complex temporal questions need additional predicates.

Source interpretation remains separate from inference over the resulting model. The automatic adapter models 1.49% of development units and 1.36% of test units. Its recovered cross-claim links occur in calibration articles. A model-assisted audit accepts 54 of sixty sampled event interpretations. Eight accepted events omit paragraph-level ends; five additional endpoint interpretations remain uncertain. Fallback excerpts remain available without certified temporal support. The separate replacement challenge uses reviewed annotations and boundary queries. Its twenty paragraphs measure mechanism behavior, rather than representative retrieval accuracy. Both reviewers are model-assisted, and the recorded source exclusions cannot establish complete absence of earlier exposure.

The answer evaluation uses one small quantized model and sixty primary questions. Structured decoding controls response syntax without establishing semantic correctness. The separate eleven-question diagnostic retains a historical cohort selected for certified context disagreement. TimeQA anchors can be aliases, headings, or mentions outside the relevant period. Strict annotated-answer metrics differ from expert judgments of semantic correctness.

The evaluated questions have previously inspected outcomes. Source-audit findings guide the automatic adapter, which remains frozen throughout the reported evaluation. The earlier TimeQA human and template test tracks overlap. TempLAMA annual labels can combine concurrent values with within-year succession. The matched Graphiti comparison isolates its temporal policy kernel under supplied candidates. It does not evaluate Graphiti’s semantic extraction model. The

release preserves source licenses, raw results, and each experiment’s scope.

Ethics Statement

The experiments use published historical passages and benchmark questions. The system preserves original source spans for inspection. Incorrect source interpretations can misstate a person’s role or tenure. The certificates describe uncertainty in the supplied records and do not establish historical truth. The release preserves dataset attribution and model licenses. Source assessments use model-assisted review and are identified as such.

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A.1 Point support

Lemma 1 (Exact point support). *For every date t ,*

$$P_{[t,t]}(c) \iff \ell_c \leq t < p_c, \quad (11)$$

$$H_{[t,t]}(c) \iff u_c \leq t \wedge (t \leq \ell_c \vee t < r_c). \quad (12)$$

Proof. If $t < \ell_c$, no assignment makes c active. If $t \geq p_c$, a witness satisfies $\ell_d > u_c$ and $u_d \leq t$. It occurs strictly after c and by t in every assignment. Thus possible support requires $\ell_c \leq t < p_c$.

Conversely, assume these inequalities. Choose $x_c = \min(u_c, t)$. For each accepted witness, choose $x_d \leq x_c$ when its bounds permit that choice. Otherwise, $\ell_d > x_c$. If $x_c = t$, this witness occurs after t . If $x_c = u_c < t$, the definition of p_c gives $u_d > t$. Choose the witness after t in this case. Independent date bounds permit all these choices simultaneously. No witness then retires c by t .

For guaranteed point support, every assignment has $x_c \leq t$ exactly when $u_c \leq t$. Assume this condition. A witness can retire c by t exactly when some assignment satisfies $x_c < x_d \leq t$. Choose $x_c = \ell_c$. A suitable witness date exists exactly when $t > \ell_c$, $\ell_d \leq t$, and $u_d > \ell_c$. The minimum eligible ℓ_d is r_c . Possible retirement therefore requires $t > \ell_c$ and $r_c \leq t$. Its complement gives the stated condition. $\square$

The equality term is necessary. Take $[\ell_c, u_c] = [0, 0]$ and $[\ell_d, u_d] = [0, 1]$ with $G(d, c)$ true. Claim c is certainly active at zero. A strictly later replacement cannot have occurred by zero. It is not certainly active at any later date. An ordinary half-open interval cannot represent this singleton. The implementation retains the exact comparisons from the lemma.

A.2 Window support

Proof of Theorem 1. Possible point support occupies $[\ell_c, p_c)$ by Lemma 1. This set intersects $[a, b]$ exactly when $\ell_c \leq b$ and $a < p_c$. The two existential quantifiers commute, proving Equation 5.

For guaranteed support, first suppose $u_c > b$. The assignment $x_c = u_c$ gives no support within the window. Thus $u_c \leq b$ is necessary. Now assume that inequality. Every assignment with $x_c \geq a$ supplies support at x_c itself. For an earlier occurrence, support fails exactly when a direct witness retires c by a . Such an assignment exists exactly when some witness satisfies $x_c < x_d \leq a$. The preceding lemma shows this is equivalent to

$a > \ell_c$ and $r_c \leq a$. Taking its complement proves Equation 6. $\square$

For each world, support means existence of a suitable time. The suitable time can differ across worlds. Consequently, a window can have guaranteed support even when no point has certain support. This distinction remains after exact text matching.

A.3 Sharpness and refinement

The possible filter retains precisely the union of world-specific support sets. Any smaller filter omits a supported claim in some world. The guaranteed filter retains precisely their intersection. Any larger filter includes an unsupported claim in some world. These statements establish the claimed minimal and maximal filters. They concern each retained claim's support, rather than one jointly optimal context.

Narrowing date bounds replaces Ω with a subset $\Omega' \subseteq \Omega$. Existential support can only disappear when its assignment set shrinks. Universal support can only appear. This proves refinement monotonicity without assumptions about gate transitivity. For correlated dates, a product approximation instead gives conservative bounds. It can enlarge possible support and reduce guaranteed support.

A.4 Local gate influence

Suppose gates G and H differ on directed pairs with target set U . Every claim outside U uses identical terms in both boundary minima. Its support status therefore remains unchanged for every query window. Thus either mask difference lies within U , where $|U| \leq h$.

Fix a complete candidate ranking and deterministic tie handling. Changing one eligibility bit alters at most one selected item and one replacement. Apply this argument successively to the targets in U . The resulting context difference is at most $2|U|$. Each context also contains at most k items. The difference is therefore at most $2 \min(k, |U|) \leq 2 \min(k, h)$.

A directed gate edit differs from a symmetric matching edit. A symmetric edit can change both directions and affect two target claims. Overlapping date bounds do not define one fixed older endpoint. A date edit can also affect several target certificates. The local bound does not treat either edit as one directed decision.

A.5 Exact dates and compilation cost

When $\ell_c = u_c = x_c$ for every claim, both boundary minima select the earliest strictly later direct witness. The possible and guaranteed masks then coincide for every window. For a point query, they equal the exact-date earliest-witness policy.

The reference compiler evaluates all $n(n-1)$ candidate directed gates. The certificates require this scan or a proved complete candidate index. Two stored witnesses alone cannot verify that no omitted witness changes a boundary. If accepted pairs are already available, two minima require $O(n+m)$ time for m pairs. The output contains at most $2n$ witness records. Every point or window decision takes $O(1)$ time per claim. A full mask takes $O(n)$ time. A complete text index can reduce gate evaluations without changing these results. No subquadratic guarantee is asserted for candidate discovery.

The worked certificate in Section 4 proves worst-case witness minimality.

B Source and Evaluation Details

B.1 TempLAMA source reconstruction

We recover the complete official test release and a public alias mirror. Both contain the same 34,963 records and the same answer identifiers. The normalizer verifies record IDs, query text, first answers, and full answer sets. The source manifest stores byte hashes for both inputs. It preserves all 45,866 answer observations.

A source key combines the subject identifier and relation identifier. The revised evaluation reserves all 300 previously used keys for legacy development. A deterministic key split assigns 764 additional keys to development and 2,993 to test. No key appears in two partitions. The test contains 25,752 yearly records and 33,908 answer observations. All thresholds and baseline choices use development records.

A year is a benchmark observation bucket. It is not a publication timestamp or an independently observed update date. Multiple answers can reflect within-year succession or concurrent values. An answer absent from a later label set is not independently verified as retired. The source audit consequently reports changes in label sets. It does not convert these changes into new occurrence bounds.

Textual names do not always identify one source entity. Six parsed slots merge 13 source entities. Fifty identical question-year groups cover 105

source rows. Sixty-one of those rows occur in the held-out test partition. The evaluation keeps these cases. It does not expose hidden entity identifiers to the text parser.

B.2 TimeQA provenance and positive labels

The source repository is pinned to commit 38b0598; the manifest records its full identifier. The release includes immutable download URLs, SHA256 hashes, and Git blob checks. The normalizer preserves each original paragraph and its character offsets. Repeated relation histories reuse the same paragraph identifier. The test corpus contains 33,493 distinct paragraphs.

The test annotations contain 384 empty answers with zero-length spans. The human subset contains 159 such annotations. The positive retrieval tracks exclude these placeholders. They retain 2,613 template-rendered questions and 830 original human questions. The human track is a subset of the larger test collection. It is not an independent second benchmark.

Eight articles occur in both the published development and test collections. Removing them before development produces 2,651 nonempty development questions. The natural-passage controls have fixed rules and no fitted parameters. The source audit records all removals and overlapping article paths.

A matched string span identifies the benchmark’s retrieval anchor. Some anchors appear in headings or in passages about another period. Aliases can also create several labels for one entity. The evaluation therefore names its metrics annotated passage hit and annotated span coverage. Entailment requires a separate source-support judgment.

B.3 Source-grounded examples

The Chicago Stadium paragraph gives two overlapping tenancies. The Blackhawks’ stated period is 1929–1994. The Bulls’ stated period is 1967–1994. Both cover the question’s period, 1967–1988. This example supports coexistence directly from the passage.

The News Quiz paragraph describes several hosts between 1977 and 1995. It includes Barry Norman, Barry Took, and Simon Hoggart. Barry Took returns after an intervening host. This example illustrates range coverage and recurrence. Its multiple answers do not establish simultaneous hosts.

A Ruud Lubbers paragraph mixes month-level and day-level employment dates. The associated question uses year-level bounds. Other passages use expressions such as *before 1516* and *around 1713*. These examples motivate preserving source precision. They do not supply numerical uncertainty widths by themselves. The release records their original text and exact offsets.

B.4 The Warburg source example

A Warburg Institute paragraph lists Henri Frankfort, Gertrud Bing, and Ernst Gombrich as successive directors (Chen et al., 2021). The paragraph dates their starts to 1949, 1955, and 1959, respectively. A deterministic clause adapter recovers these three starts and their two direct succession links from the paragraph. The passage separately dates Bing’s earlier organizational membership to 1922. That membership date does not start her directorship.

Consider the illustrative period January 1954 through June 1955. Frankfort has guaranteed support somewhere within that period. Bing has possible support because her appointment might precede July. Gombrich has no support within the period. Rank Frankfort first and Bing second. A one-item context always contains Frankfort. A two-item context changes when Bing’s start moves from January to December 1955. Both assignments satisfy the source’s year-level precision. The same possible pool thus yields different stability decisions at different budgets. The source is natural; the query, ranking, and assignments illustrate the theorem.

C Archived Results and Source Support

The supplement preserves previous generations, metrics, and source files. These records use a selected-answer protocol. Their answer metric checks normalized target containment. It does not require the complete answer set. The new source-only reader runs use a separate complete-answer protocol.

The previous TempLAMA builder kept the first listed answer. Its 300 keys contain 447 changes in that selected answer. In 153 cases, the previous answer remains in the new full answer set. The revised scorer instead accepts every official answer identifier. The annual controls use this complete-source protocol.

The archived extractive reader accesses an annotated value field. The archived historical evaluator also uses a gold anchor. Those outputs remain re-

producible as historical records. They are separate from the source-only controls in the main body.

The RealProse source audit identifies 17 excerpts without their assigned subject-role statement. It identifies 33 excerpts without the assigned start year. One excerpt gives a different start year. These findings concern the supplied passages, rather than the underlying facts. TimeQA provides the primary natural-text retrieval source in this revision.

D Reproduction

The release includes the paper sources, vector figures, compilers, corpus normalizers, predictions, and result cards. A verification entry point reruns the compiler checks and reconstructs the paper tables. Longer entry points rerun full-source and natural-passage retrieval. Input and prediction hashes identify the exact evaluated artifacts. The Graphiti policy adapter includes the inspected upstream code segments and their license. It isolates temporal invalidation under supplied candidates. It does not reproduce Graphiti’s semantic extraction model.

E Explicit Lifetimes and Fixed Fallbacks

Let claim c start at $x_c \in [\ell_c, u_c]$ and end at $z_c \in [v_c, w_c]$. Assume $u_c < v_c$ and independent start and end dates. Its explicit lifetime is $[x_c, z_c]$, additionally shortened by accepted direct replacements. Introduce a hidden event e_c with bounds $[v_c, w_c]$ and the pair $e_c \rightarrow c$. Every world has $x_c < z_c$. The hidden event retires c exactly when the explicit end occurs. It therefore preserves activity in every world. The support boundaries update to

$$p'_c = \min(p_c, w_c), \quad r'_c = \min(r_c, v_c).$$

The same support formulas apply, with at most two witness records per claim. Hidden end events never enter a reader’s context.

The strict separation condition is necessary for this independent-date reduction. For start bounds $[0, 2]$ and end bounds $[1, 3]$, the product permits start 2 and end 1. A valid lifetime excludes that assignment, but direct replacement ignores an earlier event. The adapter therefore leaves overlapping start–end bounds unmodeled. It also retains approximate dates without justified finite bounds as unmodeled text.

Let F contain fixed fallback items and S_x contain the modeled support set. The operational mask

is $S_x \cup F$. Its union over worlds is $P \cup F$, and its intersection is $H \cup F$. The context test therefore compares $\text{Top}_k(P \cup F)$ with $\text{Top}_k(H \cup F)$. Fallback eligibility records a policy choice and never receives the guaranteed-support tag.

Additional source constraints can restrict the independent date rectangle. Possible support then remains an outer bound, while guaranteed support remains an inner bound. Equal endpoint contexts still certify stability under the extra constraints. Different endpoint contexts need not establish instability within that smaller family. Counterworld exactness uses the stated independent-date model.

F Constructive Context Uncertainty

Fix a query $Q = [a, b]$, a complete strict ranking, and a budget k . Let P and H be the exact possible and guaranteed support sets. Let F contain any source items retained under a fixed fallback policy. Fallback items have unknown temporal support. They enter every operational retrieval mask. Define

$$U = \text{Top}_k(P \cup F) \setminus (H \cup F).$$

The exact context theorem makes U empty precisely when the context is stable.

Construct two witnessing timelines. Choose $c \in U$. Fewer than k items in $P \cup F$ outrank c . Every world supporting c must therefore select it. Set $t = \max(a, \ell_c)$ and $x_c = \min(u_c, t)$. Possible support gives $t \leq b$ and $t < p_c$. For every nonpivot event d , assign

$$x_d = \begin{cases} \ell_d, & \ell_d \leq x_c, \\ u_d, & \ell_d > x_c. \end{cases}$$

The first case places d no later than c . For every accepted witness, the second case places d after t . Indeed, if $x_c = t$, then $u_d \geq \ell_d > t$. Otherwise, $x_c = u_c < t$ and $\ell_d > u_c$. The definition of p_c then gives $u_d \geq p_c > t$. This rule also makes valid choices for all events outside the incoming witness set. Those choices cannot affect whether c is active. The assignments respect all independent date intervals. Claim c is active at t , so this world selects it.

For a second world, first consider $u_c > b$. Set $x_c = u_c$ and every other date to its lower bound. The claim starts after the query and cannot enter its context. Otherwise, failed guaranteed support gives $a > \ell_c$ and $r_c \leq a$. Choose the stored witness d for r_c . Set $x_c = \ell_c$ and $x_d = \min(u_d, a)$. The

witness conditions give $\ell_d \leq x_d \leq u_d$ and $\ell_c < x_d \leq a$. Thus d retires c by the query start. Set every remaining date to its lower bound. This world cannot select c . The two contexts differ.

Let n count modeled events and N count ranked excerpts, including fixed fallback items. The construction scans the event bounds and pivot certificate in $O(n)$ time. It returns $O(n)$ date values and needs no retained pair list. The pivot's rank condition already proves that these assignments yield different contexts. Optional replay scans the accepted pairs and ranking in $O(n + m + N)$ time. Explicit end events enter the same construction but remain hidden from the context. Direct context replay retains the accepted pairs and uses $O(n + m + N)$ audit state. The compact $O(n)$ certificate output stores support boundaries, rather than the full graph.

Every current pivot must be resolved. Narrow the date intervals without emptying any interval. Keep all other inputs fixed. Let P' and H' denote the refined support sets. Then $P' \subseteq P$ and $H \subseteq H'$. Take $c \in U$ that remains in P' . It still has fewer than k possible items above it. Thus $c \in \text{Top}_k(P' \cup F)$. If the refined context is stable, the exact context theorem requires $c \in H' \cup F$. Since $c \notin F$, it must belong to H' . Therefore, each current pivot becomes guaranteed or impossible.

This condition is necessary but not sufficient. For example, rank two claims c, d with independent start bounds $[0, 2]$. Use no replacement pairs, point query 1, and budget 1. Initially, the frontier is $\{c\}$. Refining c to start 2 resolves that pivot but exposes frontier $\{d\}$. The result therefore gives no minimum number of date refinements.

G Source Pipeline and Run Provenance

G.1 Calibration, revision, and source interpretation

Removing eight development–test article overlaps leaves 2,651 eligible development questions. A fixed hash selects 176 questions from 50 articles for the ranking pilot. The initial remainder contains 2,475 questions from 679 articles. Earlier source inspections identify 35 further articles for exclusion before their first validation outcomes. Removing their 127 questions gives the 2,348-question template track. The human track retains 830 positive questions from the published test collection. Both tracks had inspected outcomes before the final source revision. The source revision preserves their

questions, ranking, budgets, parser, and metrics. The structured reader replication changes decoding settings while retaining those source contexts. The release identifies these runs as post-audit revision comparisons.

Initial source calibration covers 86 articles, including 85 with eligible development questions. It combines 50 pilot articles with 39 previously identified source paths, with three overlaps. Ranking selection uses only the 176-question pilot. An independent feature audit changes query metadata and remaps source identifiers without changing selections. Ranking accepts the question string alone. BM25 uses $k_1 = 1.5$ and $b = 0.75$. Word counts split on whitespace; titles and temporal metadata add prompt text. The title inventory comes from the source corpus. Every ranking retains every corpus unit exactly once.

A year maps to January 1 through December 31 on the Gregorian ordinal axis. A month maps to its actual first and last dates. A fully stated day gives a singleton. Tenures keep separate uncertain start and end events. The final adapter requires local subjects, preserves abbreviation boundaries, checks date ownership, and links uniquely resolved paragraph-level ends. Recognized bounded states with unresolved endings remain fallback text. No answer labels guide these repairs. Twenty-seven source checks pass before the final evaluation. Five inspected calendar-end cases receive explicit endpoints. Six inspected duration cases remain fallback text because their endings cannot enter the independent-bound model. These eleven cases guide development and do not estimate extraction accuracy.

All source words survive the modeled/fallback partition. Independent checks find no overlapping units and no rejected pair endpoints. The development and test pools contain 33,005 and 34,082 units, respectively. Their modeled claim counts are 493 and 464, with 292 and 238 explicit ends. Sixteen paragraph-level end links enter the two pools. The development corpus contains six succession starts and three directed pairs. The test corpus contains two succession starts and no pairs. All three pairs occur in previously inspected calibration articles. The Warburg example uses automatic extraction; its question period and ranking are illustrative. Its selected narrative dates Bing’s appointment to 1955, while another paragraph lists 1954. The example compiles the stated narrative without resolving that cross-paragraph discrepancy.

Earlier source audit. The preceding frozen version was assessed on 60 development claims outside its calibration articles. That model-assisted audit accepted 52 events, rejected five, and marked three uncertain. Eleven accepted starts omitted terminations elsewhere in their paragraphs. Twenty-seven of its 29 explicit-end records had accepted event grounding. The final adapter addresses those findings through the five source-attachment checks described above. The earlier judgments remain unchanged in the release.

Recorded source audit. After the adapter freeze, a fixed hash selects 60 claims from 55 articles. The sample contains twenty development claims and forty test claims. It excludes 277 documented or identifiable exposure articles. The complete membership of a lost earlier audit is unavailable. We cannot rule out prior exposure for every article. Reviewers read complete source paragraphs without benchmark questions, labels, retrieval scores, or reader outputs. They accept 54 core interpretations, reject four, and mark two uncertain. All 408 checked spans match the source characters exactly. Twenty-six of 27 supplied ends are accepted; one is rejected. Core correctness and lifecycle completeness receive separate judgments. Eight accepted events omit explicit paragraph-level ends; five further endings have uncertain applicability. These model-assisted judgments are neither human annotations nor extraction-recall estimates. The samples differ, so the earlier 52/60 and current 54/60 are not paired improvement estimates. No judgment changes the final frozen adapter.

G.2 Earlier reader protocol

The earlier model is Qwen2.5-3B-Instruct, Q4_K_M, at revision 7dabda4d. The release records the full revision and weight checksum. The runtime is llama.cpp b11435 at commit 43fe9c6. The model uses the Qwen Research License; the runtime uses MIT. Generation weights are excluded from the archive. Pinned download scripts restore the official artifacts.

Prompts contain the visible question, original excerpts, source titles, date bounds, and temporal status. Modeled excerpts also carry source-derived subject and relation labels. Those labels preserve role context for short succession clauses. No extracted answer-value field is added to the prompt metadata. Temperature is zero, seed is 1729, and the output limit is 96 tokens. Prompt caching and

JSON grammar enforcement are disabled. A tokenizer check verifies that every request fits the configured context. Resource-only amendments precede completed question-answer outputs and preserve the original freezes and logs. The earlier runtime uses four threads, a 2,048-token context, and polling disabled. Its batch and microbatch sizes are both 512.

The parser accepts a complete JSON object or a single JSON code fence. It rejects narrative repair, invalid field types, invalid excerpt indices, and truncated outputs. Failures receive empty answer lists and remain in every denominator. Normalization applies Unicode NFKC, case folding, and word-token extraction. It preserves articles and compares deduplicated complete answer sets. Strict set precision, recall, F1, and exact set accuracy use those normalized sets. Token overlap remains a secondary measure.

The primary sample contains 60 questions from 55 articles. Its 600 method conditions reduce to 194 distinct prompts. The diagnostic retains eleven questions selected by the preceding source version without answer labels or generated answers. That version selected different possible and guaranteed contexts, up to twenty-four questions. The revised contexts still differ on seven of those eleven questions. All eleven remain evaluated, yielding 110 conditions and forty distinct prompts. They come from nine articles, four shared with the primary sample. The cohorts have no question overlap and no shared requests. Of 234 earlier requests, 166 exactly match archived requests under the same model and execution settings. The other 68 requests receive new generations. Reuse checks include full messages, output limits, seed, stop conditions, model hashes, and runtime settings. Full request hashes, raw responses, timing, and per-request provenance support independent verification.

Complete reader outcomes. Table 4 reports both cohorts separately. The diagnostic guaranteed-minus-possible F1 difference is 9.09 points, with article-cluster interval [0.00, 30.00]. All 45 paired method contrasts remain in each cohort’s saved summary. The primary temporal contrasts change no answer sets, including all 37 pairs with valid responses on both sides. The diagnostic possible–guaranteed contrast changes three answer sets among seven pairs with two valid responses. One further change involves a response-format failure.

Method	Primary		Diagnostic	
	EM	F1	EM	F1
Original BM25	16.67	16.67	9.09	9.09
Original BM25 + year	18.33	18.33	9.09	9.09
Title-aware ranking	21.67	21.67	9.09	9.09
+ earliest completion	20.00	20.00	18.18	18.18
+ midpoint completion	20.00	20.00	9.09	9.09
+ latest completion	20.00	20.00	9.09	9.09
+ interval control	20.00	20.00	9.09	9.09
+ possible support	20.00	20.00	9.09	9.09
+ guaranteed support	20.00	20.00	18.18	18.18
Title-aware + year	21.67	21.67	9.09	15.15

Table 4: Earlier unconstrained decoding. Reader exact match (EM) and strict answer-set F1 (%). Primary: 60 questions and 600 conditions. Diagnostic: 11 questions and 110 conditions. Unique generation parse failures: 69/194 and 11/40, respectively. Failures remain in both denominators.

Earliest–latest completion gives the same counts and changed questions. Only the Maher change increases strict F1. The other valid changes concern Atlético Madrid and Helen Clark; both retain zero strict F1. This diagnostic retains its earlier selection while evaluating the revised source contexts.

Across 234 distinct earlier requests, eighty responses fail the frozen parser. These comprise 58 JSON-format failures, four invalid citation lists, and eighteen length-limit responses. Every failed response remains scored with an empty answer set. The primary ranking comparison changes twenty answer sets against original BM25 with recency. Five changes occur among 24 pairs with two valid responses; fifteen involve a parsing failure. Separate descriptive audits report this breakdown for every method pair without changing the frozen scores. The release retains 238 earlier executions and 68 new executions, totaling 306 archived executions across both versions.

G.3 Ranking portability and selection cost

A secondary analysis applies the certificate to all six ranking variants considered during calibration. The revised protocol precedes its mechanism analysis and preserves the six variants from the earlier ranking study. It reads no answer labels and does not replace the selected main ranking. Each variant has its own modeled-context denominator. The release also reports fixed main and union cohorts for direct comparisons. The lazy selector agrees with full-mask evaluation for every parsed query. Temporal-test counts include possible and guaranteed predicate calls. Fallback excerpts require no

Ranking	Template stable/modelled	Human stable/modelled	Template tests	Human tests
Global BM25	285/354	86/102	2.64	2.34
Global BM25 + year	443/534	138/164	2.72	2.39
Title BM25	241/298	76/86	2.66	2.41
Title-aware	189/243	56/66	2.63	2.44
Title BM25 + year	327/396	109/126	2.67	2.42
Title-aware + year	281/343	102/120	2.56	2.40

Table 5: Secondary certificate analysis across all six prespecified rankings. Each stability denominator contains modeled evidence under that ranking. Tests count temporal predicate calls within that modeled cohort.

predicate call. The timing microbenchmarks exclude extraction, compilation, relevance ranking, rendering, and answer generation. Concurrent CPU workloads affect their absolute timings. The main text therefore reports operation counts.

G.4 Research assistance and verification

AI assistants contributed to code development, theoretical derivations, manuscript drafting, figure construction, and citation checking. The source interpretation audits are model-assisted judgments. The release contains independent checks, complete execution records, and a submission-form disclosure covering these uses.

G.5 Earlier reader and extraction calibration

Earlier Qwen2.5-1.5B-Instruct runs remain available under their original frozen settings. A 24-question pilot requires 52 generations for 192 conditions. A nine-question diagnostic adds 22 generations and reuses six. The cohorts overlap on two questions, giving 31 distinct questions and 74 actual generations. All methods reach 31.94% strict set F1 on the pilot. The earlier diagnostic changes four answer sets between earliest and latest completion, without improving aggregate strict F1.

Two exploratory learned extractors each process six calibration paragraphs before stopping. The first returns 41 records, all missing required source quotes. A schema-constrained variant returns 46 records; 45 fail literal grounding. The remaining record has an incorrect holder-date association. Neither extractor supplies claims to the reported confirmation runs. Partial manifests retain all completed outputs, settings, and stopping decisions.

An earlier deterministic reproduction correction sorts BM25 token accumulation and stops after five accepted units. Every one of 20,192 archived method contexts remains unchanged. The original runner and equivalence records are preserved.

Source-label property	Count	Rate (95% CI)
Multiple answers	6,498/25,752	25.2 [24.2, 26.2]
Old first retained	1,456/3,980	36.6 [35.1, 38.0]
Any answer retained	6,050/6,961	86.9 [86.0, 87.8]

Table 6: Answer-set properties on held-out source keys. The last two rows condition on first-answer changes and complete-set changes, respectively. Rates and intervals are percentages.

Policy	H@1 ↑	R@5 ↑	S@5 ↓
BM25	64.64	80.84	45.89
BM25 + recent ties	84.04	90.10	40.50
Age weighting	99.81	99.98	24.07
Date-first ranking	99.81	99.98	0.00
Original predicate	99.37	99.77	0.77
Graphiti kernel	99.78	99.93	0.00
Query latest-value	99.89	86.37	0.00
Query latest-batch	99.89	99.98	0.00

Table 7: Annual source-label retrieval on 25,752 held-out questions. All values are percentages. H@1: first-result accuracy; R@5: answer recall; S@5: target-label stale exposure.

G.6 Complete annual controls

The annual split and complete-source labels match the archived experiment. Latest-value selection retains 99.89% first-result accuracy but achieves 86.37% complete answer recall. Latest-batch selection reaches 99.98% recall. The suppression difference is 13.63 points, with key-cluster interval [13.06, 14.21]. Graphiti’s matched policy kernel and direct-first invalidation produce identical endpoints across 45,866 observations. These retained results require no repeated model inference.

G.7 Earlier natural paragraph controls

These paragraph runs use the published test passages and positive annotations. Their human-question results were inspected before the source-unit experiments. Paragraph and source-unit runs have different retrieval units and denominators.

Policy	Human		Templates	
	H@5	R@5	H@5	R@5
BM25	45.66	45.04	32.49	31.75
Latest year	54.70	54.02	51.86	50.89
Year envelope	49.16	48.33	34.44	33.67
Explicit range	44.10	43.53	32.11	31.39
Range support	44.58	44.00	38.50	37.63

Table 8: TimeQA paragraph retrieval percentages. H@5 measures annotated-passages coverage; R@5 measures annotated-span recall. The tracks contain 830 human questions and 2,613 template questions.

Certificate cohort	Template	Human
Questions	2,348	830
Parsed query periods	2,296	813
Contexts with modeled evidence	243	66
Certified stable contexts	189	56
Unstable contexts	54	10
Stable within modeled group	77.78%	84.85%

Table 9: Stability for the frozen title-aware ranking. The modeled group contains parsed questions with at least one modeled excerpt in the possible context. All-fallback contexts have a separate denominator.

G.8 Complete matched retrieval and budget results

Tables 9–11 collect every cohort, budget, and policy from the automatic source-unit evaluation.

H Hardness for Input Context Budgets

We reduce Set Cover (Karp, 1972) to possible top- k membership. The context budget k is part of the decision input. Take a universe E , sets $S_1, \dots, S_s$, and budget $1 \leq K \leq s$. Create variable claim v_j with date interval $[1, 3]$ for every set. Create $K + 1$ copies of each element claim, all dated exactly 0. Create candidate c dated exactly 0. Accept $v_j \rightarrow e$ for every copy of each $e \in S_j$. There are no other accepted pairs. The gate is bipartite, with one edge layer from variables to element copies.

Rank all variables and element copies above c . Set query time 2 and budget $k = K + 1$. Candidate c has guaranteed support. A variable is active exactly when its date is at most 2. Every active variable retires its associated element copies by the query time. An inactive variable starts after the query and cannot retire any copy by that time.

Given a cover of size at most K , assign its variables date 1 and others date 3. Every element copy is retired. At most K higher-ranked variables remain active, so the context includes c . Conversely, suppose a world includes c . At most K higher-

Excerpt budget k	1	3	5	10	20
Template ($n = 243$)	90.9	79.4	77.8	77.0	76.5
Human ($n = 66$)	97.0	90.9	84.8	83.3	78.8

Table 10: Stable contexts (%) as the unit budget grows. Each row retains the same modeled cohort selected at $k = 5$. Selection precedes word truncation.

Method	Template	Human
Original BM25	37.63	44.80
Original BM25 + latest year	53.27	53.11
Title-aware ranking	67.61	58.11
+ earliest completion	67.87	58.47
+ midpoint completion	67.76	58.35
+ latest completion	67.78	58.35
+ interval control	67.89	58.47
+ possible support	67.89	58.47
+ guaranteed support	67.76	58.35
Title-aware + latest year	66.55	64.26

Table 11: Mean question-level annotated span recall (%) within five excerpts and 768 source-text words. Temporal controls share the title-aware ranking. Latest-year controls change only ranking.

ranked claims are active. The variables dated at most 2 form a collection of at most K sets. If they miss an element, its $K + 1$ active copies exclude c . They must therefore cover E . The construction has polynomial size and proves NP-hardness. It does not establish hardness for a fixed budget such as $k = 5$.

The result also holds when every start occurs by the query time. For this variant, use variable bounds $[0, 2]$, element dates 1, and candidate date 2. Add a clock at date 1, ranked below the candidate, with a pair targeting each variable. Use the same variable-to-element pairs and query time 2. Variables before 1 retire under the clock. Variables at or after 1 remain active, but only those after 1 retire element copies. A cover assigns its variables 2 and all others 0. Conversely, a selected candidate permits at most K active variables. Those after 1 must cover every element, or $K + 1$ active copies exclude the candidate. Variables exactly at 1 add cost without covering an element. This variant has an acyclic gate with two edge layers.

This hardness concerns arbitrary possible context membership. It does not contradict the linear exact stability test. An unstable possible context already supplies a pivot with fewer than k possible items above it. That rank condition makes its supporting world sufficient to demonstrate a context change.

I Exact Queries Under Source Constraints

Feasible calendar dates. This extension represents event dates as integers. Let $\mathcal{C}$ contain source constraints of the form $x_i - x_j \leq v$. The feasible family is

$$\Omega_{\mathcal{C}} = \{x : \ell_i \leq x_i \leq u_i \text{ for all } i, x \models \mathcal{C}\}.$$

The implementation rejects an empty family before evaluating support. A known order $x_i < x_j$ becomes $x_i - x_j \leq -1$. An explicit end z_c requires $x_c - z_c \leq -1$ and the hidden replacement pair $z_c \rightarrow c$. This constraint permits overlapping start and end bounds while excluding invalid lifetimes. End events remain hidden from the reader's context.

Three query predicates. Write $D(c) = \{d : G(d, c) = 1\}$ for the accepted direct replacements of c . For query $[a, b]$, define overlap support, throughout support, and appointment support by

$$S_x^{\text{overlap}}(c, [a, b]) \iff x_c \leq b \wedge \Phi_c(a), \quad (13)$$

$$S_x^{\text{throughout}}(c, [a, b]) \iff x_c \leq a \wedge \Phi_c(b), \quad (14)$$

$$S_x^{\text{appointed}}(c, [a, b]) \iff a \leq x_c \leq b, \quad (15)$$

where

$$\Phi_c(t) = \bigwedge_{d \in D(c)} (x_d \leq x_c \vee x_d \geq t + 1).$$

Appointment support concerns the start event, independently of later replacement. Throughout support respects half-open activity. A replacement at b therefore invalidates support throughout $[a, b]$.

For overlap, the first possible supporting date is $\max(a, x_c)$. Support requires $x_c \leq b$ and no replacement with $x_c < x_d \leq a$. Negating the replacement condition gives Equation (13). Throughout additionally requires a start by a and excludes replacement through b . This proves Equation (14).

Exact feasibility oracle. Possible support asks whether one feasible assignment satisfies the selected predicate. The oracle branches over its replacement disjunctions. Every branch contains only difference constraints. An infeasible conjunction prunes that branch. A successful branch returns a supporting timeline.

Guaranteed support holds exactly when the predicate's negation has no feasible assignment. The overlap negation is

$$x_c \geq b + 1 \quad \vee \quad \bigvee_{d \in D(c)} (x_c + 1 \leq x_d \leq a).$$

For throughout, replace $b + 1$ by $a + 1$ and the final a by b . Appointment uses the alternatives $x_c \leq a - 1$ and $x_c \geq b + 1$. Each feasible branch returns an excluding timeline.

The implementation solves each conjunction through weighted graph relaxation. A negative cycle establishes infeasibility. Otherwise, final distances satisfy the constraints. Subtracting the zero-node distance preserves every difference and yields a valid timeline. Integer edge weights produce integer dates. Thus the oracle returns exact support decisions and valid witness assignments.

Let $d = |D(c)|$, and let e count base constraints, including bounds. Each feasibility check takes $O(n(e + d))$ time. Possible support can require $O(2^d)$ checks. Guaranteed overlap or throughout support needs at most $d + 1$ checks. The constrained oracle therefore has a different cost from the independent compiler.

Stability does not require independent dates. Take any nonempty timeline family Ω and one fixed query predicate. Let S_x denote its support set in timeline x . Define $P = \bigcup_{x \in \Omega} S_x$ and $H = \bigcap_{x \in \Omega} S_x$. Under a fixed complete ranking, the context is stable exactly when

$$\text{Top}_k(P) = \text{Top}_k(H).$$

To prove sufficiency, every selected possible item is guaranteed and outranks every remaining possible item. For necessity, choose $c \in \text{Top}_k(P) \setminus H$. Some timeline supports c , while another does not. Fewer than k possible items outrank c . The supporting timeline therefore selects it, and the excluding timeline cannot. The proof also permits fixed fallback items in every operational support set.

Screen first, then solve unresolved candidates. Let P_0, H_0 denote independent-product overlap support. Because $\Omega_{\mathcal{C}}$ restricts that product,

$$H_0 \subseteq H \subseteq P \subseteq P_0.$$

Equal endpoint contexts immediately certify the constrained family. This screen uses no query-level feasibility calls. The model's nonemptiness is checked once before querying.

Otherwise, scan candidates in their fixed relevance order. Skip claims outside P_0 and accept claims in H_0 as guaranteed. Query exact constrained support for the remaining candidates until k possible claims are selected. Report stability when every selected claim is guaranteed. The inclusions above validate every shortcut. Exact queries resolve every remaining candidate needed by the scan. The resulting context is therefore exactly $\text{Top}_k(P)$. Fixed fallback items extend the scan by entering every support set without a solver call. The released controlled benchmark uses only modeled claims. The general stability result proves the final decision. For instability, the first unresolved selected claim supplies two feasible witness timelines. Its rank ensures selection in its supporting timeline. Its excluding timeline cannot select it.

Controlled verification. The frozen benchmark contains 120 feasible models and 20 inconsistent models. It covers known order, valid lifetimes, and their combination. Thirty-eight explicit lifetimes have overlapping date bounds. The reference enumerates 20,846 feasible timelines and evaluates direct activity. All 100,800 support decisions and 30,240 context decisions agree. Every inconsistent model is rejected. The audit checks 68,161 returned witness assignments. All generated cases remain in the release.

For overlap queries, the hybrid agrees on all 10,080 context decisions. It screens 3,001 stable contexts without query-level solver calls. It recovers 312 additional stable contexts through exact constraints. Against lazy exact selection, it reduces claim queries from 33,180 to 13,343. Both methods stop after selecting the requested number of possible claims. Feasibility calls decrease from 141,992 to 62,952. The complete-mask oracle uses 50,400 claim queries and 218,079 feasibility calls. All 6,767 unstable contexts have two replayed timelines with different selections. These counts measure solver work without asserting an end-to-end speedup.

A ranking counterexample. Take two claims with independent start bounds $[0, 1]$ and no replacement pairs. Both claims have guaranteed support at query date 1. Any fixed ranking therefore gives a stable context for budget one. Ranking by realized start instead selects different claims in timelines $(1, 0)$ and $(0, 1)$. The support sets remain identical while the ranking changes. The fixed-ranking condition is therefore necessary for this stability

test.

The controlled grid contains 8,627 contexts certified stable under fixed ranking. Sorting instead by realized start changes 3,640 selected sets and 4,808 ordered contexts. Original ranking breaks ties between equal starts. At budgets one, three, and five, set changes occur in 2,683, 957, and zero cases, respectively. Their corresponding fixed-stable denominators are 4,604, 2,197, and 1,826. Budget five retains all eligible claims, but their order changes in 883 cases. These comparisons combine all three predicates within the same frozen grid.

The effect of query meaning. Each predicate has 16,800 claim–window decisions across the feasible models. Throughout differs from overlap on 5,840 possible-support decisions. Appointment differs on 2,989 possible-support decisions. These comparisons hold the date model and query window fixed. The predicate is an explicit input; this experiment does not evaluate automatic question interpretation.

A separate frozen wording screen finds 82 TimeQA questions containing prespecified start-event cues. Seven produce modeled contexts, and three of those contexts are unstable. The release retains all matching questions and cue strings. These lexical flags are an interpretation audit, rather than semantic labels for predicate-selection accuracy.

J Source-Derived Replacement Challenge

Selection before outcomes. We screened the saved development and test paragraphs without reading original questions or answer labels. We first excluded 332 articles from the documented exposure records. Candidates required two distinct calendar years and an explicit succession cue. The cues were *succeeded*, *successor*, *replaced*, *replacing*, *replacement*, *took over*, and *handed over*. We ordered 496 candidates by SHA-256 of a fixed seed and passage identifier. The first 128 paragraphs yielded seven accepted graphs. Before certificate evaluation, we fixed one additional panel containing the next 256 paragraphs. This panel yielded thirteen accepted graphs. The expansion responded to annotation yield, without changing eligibility rules. All 384 judgments remain available, including rejections and uncertain cases.

Two model-assisted reviews checked every accepted graph against its complete source paragraph. Reviewers divided the additional annotation work

and independently checked each other’s proposed graphs. We excluded graphs with uncertain dates, roles, or replacement relations. We also excluded named prior examples and one article inspected during the current editorial audit. Membership of an earlier lost audit remains incomplete, so this challenge does not claim fully unseen articles. The accepted panel contains seventeen test paragraphs and three development paragraphs. These reviewed graphs form a mechanism diagnostic, rather than an evaluation of automatic extraction.

Graphs and source excerpts. Each graph retains its complete paragraph, article identifier, literal offsets, and date provenance. Twenty-five edges record consecutive replacements. Eight edges connect nonconsecutive holders within complete, explicitly ordered succession lists. We tag these closure edges separately. Source succession also supplies strict order constraints between the corresponding dates. A transition stated as both departure and succession uses one shared event, avoiding an independent duplicate endpoint. Separate explicit departures retain their own bounds. All 51 calendar dates and 174 literal spans pass source checks. Displayed excerpts contain literal clauses or coherent clause segments. Excerpts can overlap because one sentence can describe both holders.

Queries and controls. Each successor contributes its earliest date, integer midpoint, and latest date as point queries. We remove duplicate dates within each graph, producing 73 queries. The predicate is overlap, with $a = b$. The fixed ranking follows start-mention offsets, with annotation order breaking ties. One fixed fallback excerpt F follows every modeled claim. We evaluate budgets $k = 1$ and $k = 5$. Both policies retain identical bounds, explicit ends, and known source orders. The interval control removes only cross-claim replacement edges. Thus the comparison isolates replacement while preserving the same feasible calendar assignments.

Exact evaluation and replay. The oracle searches every integer calendar assignment through complete difference-constraint branching, as described in Appendix I. Endpoint sampling does not supply the reference labels. The source-order comparison changes 33 selected contexts at $k = 1$ and 34 at $k = 5$. Only the first budget changes stability labels, with 42 interval-stable contexts becoming unstable. Replacement certifies 25 queries at each

budget. All 96 unstable context decisions have two feasible timelines that obey every source order and change the literal excerpts.

We also retain the independent-date comparison on the same graphs and queries. The linear compiler matches 388 exact support judgments across 194 claim-query pairs. The lazy selector matches all 146 context decisions. All 98 unstable decisions have replayed witnesses under that product model. These counts include both budgets and do not represent independent source questions.

Effect of a real source order. The Levadia paragraph dates Bondarenko’s appointment to 2001 and Rautiainen’s succession to November 2001. Their known order removes three possible-support judgments for Bondarenko. The affected queries are 30 November 2001, 1 January 2003, and 16 January 2003. This restriction resolves one additional stable query at each budget. The archive retains both comparisons, every source decision, and every replayed timeline.

K Structured Reader Replication

Unchanged evidence and scoring. The replication retains all 234 distinct requests from the earlier primary and diagnostic cohorts. Every message, question, source excerpt, and method membership remains unchanged. The ten methods produce 710 question–method conditions. The model remains Qwen2.5-3B-Instruct with the same Q4_K_M weights and pinned llama.cpp runtime. The strict parser, normalization, answer aliases, and article-cluster resampling also remain fixed. Appendix G.2 retains the earlier decoding protocol and its complete results.

Structured decoding. A JSON schema constrains the output object and citation indices. Answer strings remain unrestricted, and the schema does not prescribe the number of answers. The completion budget increases to 512 tokens, with a 4,096-token context. Temperature remains zero and the seed remains 1729. Prompt caching stays disabled. A tokenizer preflight confirms that every complete prompt fits this context. The six lexicographically smallest request identifiers form a syntax pilot. All six complete normally and pass the frozen parser before the full run proceeds. No answer scores guide this gate.

The run initially uses one four-thread CPU server. A recorded scheduling amendment adds a second

Method	Primary		Diagnostic		Valid
	EM	F1	EM	F1	
Original BM25	23.33	24.44	9.09	9.09	71/71
Original + year	25.00	25.00	18.18	18.18	71/71
Title-aware	35.00	36.11	9.09	9.09	71/71
Earliest completion	35.00	36.11	18.18	18.18	71/71
Midpoint completion	35.00	36.11	9.09	9.09	71/71
Latest completion	35.00	36.11	9.09	9.09	71/71
Interval control	35.00	36.11	9.09	9.09	71/71
Possible support	35.00	36.11	9.09	9.09	71/71
Guaranteed support	35.00	36.11	18.18	18.18	71/71
Title-aware + year	33.33	33.33	18.18	24.24	71/71

Table 12: Complete structured reader results. EM and strict answer-set F1 are percentages. Valid counts use question-method conditions. All conditions remain in the denominators.

independent server with identical settings. Completed responses remain fixed; outstanding requests alternate between workers. One interrupted request had no completed output and returns to the queue. The resulting archive contains 234 new completed responses, including the six pilot responses. The earlier 306 execution records remain unchanged. Per-request hashes identify the exact inputs, decoder settings, weights, runtime, and raw response.

Context identity and answer cardinality. Possible and guaranteed support produce identical requests for every primary question. Prompt deduplication therefore gives identical primary answers under both policies. Their diagnostic requests differ for seven questions. The primary targets contain one answer for 56 questions and two answers for four questions. Every diagnostic target contains one answer. The earlier primary decoder records no partial strict matches, which explains equality between exact match and strict F1. The new audit reports target and prediction cardinalities separately for every method.

Complete outcomes. Every response stops normally and passes the frozen parser. The longest completion contains 51 tokens. Across unique requests, 75 predictions are empty, 157 contain one answer, and two contain two answers. Empty predictions are valid abstentions and remain in every metric denominator. Table 12 reports all ten methods in both cohorts.

The title-aware primary row has 35.00% exact match and 36.11% strict F1. Its partial match concerns Vladimir Bukovsky. The prediction is *Russian*; the annotated set contains *British* and *Russian*. Strict set F1 therefore assigns 2/3 credit while exact match assigns zero. The same partial match occurs under each temporal policy.

The diagnostic possible–guaranteed comparison changes four answer sets, all with valid responses. One change gains strict F1, and three retain zero strict F1. Strict ties do not establish semantic equivalence. For Mbaye-Jacques Diop, the prediction changes from the annotated party with its abbreviation to *Socialist Party (PS)*. Both receive zero under strict normalized matching. The release records every changed question, answer, citation list, and cardinality. The diagnostic retains its historical selection and nine article clusters. Its guaranteed-minus-possible interval remains $[0.00, 30.00]$ under the prespecified article bootstrap.